

Muhammad Irfan

SENIOR COMPUTER VISION & MACHINE LEARNING ENGINEER

Grand Blanc, MI | (248) 759-3191 | mirfan.0525@gmail.com | [LinkedIn](#) | [Google Scholar](#)

U.S. Permanent Resident / U.S. Person | No visa sponsorship required

Professional Summary

Computer vision and machine learning engineer with 10+ years of research experience and current automotive industry experience developing, optimizing, and deploying deep learning systems across manufacturing and medical imaging. At General Motors, builds image and signal models and transitions proof-of-concept solutions into optimized real-time inference workloads using NVIDIA TensorRT and Jetson Thor. Research spans 2D/3D anomaly detection, segmentation, classification, image enhancement, motion analysis, multimodal imaging, data-efficient learning, and embedded vision, supported by patents and peer-reviewed publications.

Technical Skills

Computer Vision: Object detection, image classification, semantic segmentation, feature extraction, image enhancement, motion analysis, 3D imaging, camera and video analytics, multimodal and vision-language models

Edge AI & Deployment: NVIDIA Jetson Thor, TensorRT, real-time inference, embedded systems, model optimization, hardware-software integration, latency/accuracy tradeoffs, PoC-to-production deployment

Programming & Frameworks: Python, C/C++, MATLAB, PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, MONAI, NumPy, Pandas, Scikit-image

Models & Methods: CNNs, RNNs, GANs, transformers, UNet variants, SegFormer, one-shot and few-shot learning, federated learning, multimodal fusion

Platforms & Tools: AWS, Microsoft Azure, Google Cloud Platform, Git, Visual Studio, DICOM, ITK, SimpleITK, VTK, VMTK, 3D Slicer

Professional Experience

Machine Learning Engineer | General Motors R&D | Warren, MI Jan 2026 – Present

- Develop and deliver real-time computer vision and deep learning solutions from requirements and data ingestion through training, validation, optimization, system integration, and production deployment.
- Build CNN-, RNN-, and GAN-based models for complex image and signal data, including feature extraction, classification, detection, and prediction workflows.
- Optimize deep learning models with NVIDIA TensorRT and deploy inference workloads on NVIDIA Jetson Thor, balancing accuracy, latency, throughput, memory, and computational cost.
- Design reproducible model and inference pipelines that move proof-of-concept systems toward scalable operational deployment; current programs are in the deployment phase.
- Mentor 4–6 engineers, review technical designs and deliverables, resolve implementation risks, and collaborate with cross-functional engineering teams in the United States and India.
- Lead inventor on a GM U.S. provisional patent application and a separate confidential AI/ML innovation retained as a GM trade secret.

Research Assistant | SMILES Lab, University of Michigan-Flint | Flint, MI Jan 2022 – Dec 2025

- Built reproducible 3D computer vision pipelines for MRI, MRA, DSA, and clinical data, covering preprocessing, training, fine-tuning, validation, error analysis, and documentation.
- Developed detection, segmentation, and risk-prediction models using CNNs, UNet variants, SegFormer, transformers, large vision models, vision-language models, and image-text-tabular fusion.
- Applied one-shot, few-shot, and federated learning to improve generalization, data efficiency, and privacy across limited and distributed datasets.
- Produced patented and peer-reviewed computer vision research through collaborations spanning the United States, Norway, the UAE, Pakistan, and South Korea.

Research Assistant | Mixed Reality & Interaction Lab, Sejong University | Seoul, South Korea Sep 2018 – May 2021

- Designed and evaluated deep learning systems for image enhancement, dehazing, object detection, segmentation, saliency detection, motion estimation, image-quality assessment, and sensor-based activity recognition.
- Developed CNN-based perception and tracking methods for 360-degree video and immersive industrial environments, including field-of-view selection and abnormal-activity detection.
- Built experimentation and data pipelines, optimized models for accuracy and computational efficiency, and contributed to granted patents and peer-reviewed publications.

Research Assistant | Digital Image Processing Lab, Islamia College University | Peshawar, Pakistan Jun 2016 – Jul 2018

- Developed a real-time CNN-assisted autonomous-driving prototype on Raspberry Pi, integrating computer vision models with resource-constrained embedded hardware.
- Implemented and optimized lightweight motion-detection and image-enhancement models for real-time edge applications.

Teaching Assistant | Oakland University, Computer Science Department | Rochester, MI Sep 2023 – Apr 2025

- Mentored students and supported instruction in data structures and algorithms, software engineering, computer networks, C/UNIX programming, and engineering problem solving courses.

Education

PhD in Computer Science University of Michigan-Flint Flint, MI	Expected Dec 2026
Focus: Brain aneurysm detection, segmentation, and rupture prediction using multimodal imaging and clinical data.	
MS in Computer Engineering Sejong University Seoul, South Korea	Aug 2020
Thesis: Deep learning-based field-of-view selection in 360-degree video.	
BS in Computer Science Islamia College University Peshawar, Pakistan	Aug 2018
Thesis: Raspberry Pi-based CNN-assisted self-driving car.	

Patents & Intellectual Property

- General Motors confidential AI/ML invention, U.S. provisional patent application filed in 2026; lead inventor.
- General Motors confidential AI/ML innovation retained as a protected trade secret in 2026; lead inventor.
- U.S. Provisional Application No. 30275/70862P, "Adaptive Learning for Multi-Modal Intracranial Aneurysm Rupture Criticality Analysis," filed 2025; lead inventor.
- KR Patent 10-2284913 B1, "User's Field of View Selection Method and Apparatus in 360-Degree Videos," granted 2021; lead inventor.
- KR Patent 10-2168438 B1, "Method and Server for Detecting Abnormal Activity Using Drones," granted 2020; lead inventor.

SELECTED PUBLICATIONS

- Irfan, M. et al. (2025). One-Shot Learning for Generalization in Medical Image Classification Across Modalities. Computerized Medical Imaging and Graphics.
- Irfan, M. et al. (2024). Federated Fusion Learning with Attention Mechanism for Multi-Client Medical Image Analysis. Information Fusion.
- Irfan, M. et al. (2023). StrokeNet: Automated Segmentation and Rupture Risk Prediction of Intracranial Aneurysms. Computerized Medical Imaging and Graphics.
- Irfan, M. et al. (2021). Deep Learning-Based User Field of View Selection in 360-Degree Videos for Industrial Environments. IEEE Internet of Things Journal.
- Irfan, M. et al. (2021). Light-DehazeNet: Lightweight CNN for Single Image Dehazing. *IEEE Transactions on Image Processing*.
- Sajjad, M., Irfan, M. et al. (2020). Efficient and Scalable Simulation Model for Autonomous Vehicles with Economical Hardware. IEEE Transactions on Intelligent Transportation Systems.